

Q: Support Vector Machines SVM

Q13

(A)

$$6x_1 + 8x_2 - 48 = 0$$

$$P(10, 2)$$

$$Q(3, 3)$$

$$(6 \times 10) + (8 \times 2) - 48 = \neq$$

$$6 \times 3 + 8 \times 3 - 48 =$$
$$= 18 + 24 - 48 = \underline{\underline{-6}}$$

$$60 + 16 - 48 = \underline{\underline{28}}$$

P is on the positive side

Q is on the negative side

(B)

~~Distance~~

Distance:

$$\|w\| = \sqrt{6^2 + 8^2}$$

$$= \sqrt{36 + 64}$$

$$= \sqrt{100} = \underline{\underline{10}}$$

$$P: \frac{28}{\|w\|} = \frac{28}{10}$$

$$= \underline{\underline{2.8}} \text{ Distance from boundary}$$

$$Q: \frac{|-6|}{\|w\|} = \frac{|-6|}{10} = \underline{\underline{0.6}}$$

~~Distance~~

Distance from boundary

(C) the likely Support Vector between the two points would be Q since it is closer.